Outdoor Landmark Chair

Completed · Independent design and fabrication

I designed and built an oversized outdoor chair for seating and group photographs at Bucknell. The project combined SolidWorks modeling and a preliminary structural study with material selection, carpentry and a two-day construction schedule.



My SolidWorks model of the braced outdoor chair.

Design and fabrication

I developed the dimensions and braced frame in SolidWorks, selected pressure-treated lumber, and cut, assembled, sanded and finished the structure independently. The approximately 6 ft 6 in-high, 8 ft-wide and 8 ft-deep chair was built in two days for approximately $1,000.

Analysis and evidence

The archived SolidWorks plot reports approximately 1.78 MPa maximum von Mises stress for the modeled load case. The applied numerical force and full material, connection and mesh setup are unavailable. The plot documents preliminary modeling rather than a verified timber-and-joint load rating.

Outcome

The completed chair remains in fraternity use. It seats four to five people and serves as a backdrop for larger group photographs. Its related engineering principles include load paths, bracing, joint behavior and assembly sequence. This was an independent project, separate from coursework.

Evidence basis: Student-supplied engineering portfolio, 27 September 2026; associated résumé.